

|                           |                                        |                        |                       |
|---------------------------|----------------------------------------|------------------------|-----------------------|
| <b>COURSE CODE/TITLE:</b> | CPE 212   Automating Server Management | <b>SECTION:</b>        | CPE31S2               |
| <b>DATE PERFORMED:</b>    | 08/19/2026                             | <b>DATE SUBMITTED:</b> | 08/19/2026            |
| <b>NAME:</b>              | Melquisedec Iquin                      | <b>INSTRUCTOR:</b>     | Engr. Jimlord Quejado |

**ACTIVITY NO. 4.1**
**Running Elevated Ad hoc Commands**

**1. PROCEDURE OUTPUT**

What procedures did you perform?

iquin@Workstation: ~

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```

iquin@Workstation:~$ sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ansible-core python3-argcomplete python3-dnspython python3-jmespath
  python3-kerberos python3-libcloud python3-lockfile python3-ntlm-auth
  python3-packaging python3-passlib python3-requests-ntlm python3-resolvelib
  python3-selinux python3-simplejson python3-winrm python3-xmltodict
Suggested packages:
  cowsay sshpass python3-trio python3-aioquic python3-h2 python3-httpx
  python3-httpcore python-lockfile-doc
The following NEW packages will be installed:
  ansible ansible-core python3-argcomplete python3-dnspython python3-jmespath
  python3-kerberos python3-libcloud python3-lockfile python3-ntlm-auth
  python3-packaging python3-passlib python3-requests-ntlm python3-resolvelib
  python3-selinux python3-simplejson python3-winrm python3-xmltodict
0 upgraded, 17 newly installed, 0 to remove and 1 not upgraded.
Need to get 19.6 MB of archives.
After this operation, 315 MB of additional disk space will be used.
0% [Waiting for headers]S

```

```
iquin@Workstation: ~  
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter  
out any that are already installed  
  
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt  
ed now it is to install the new keys  
iquin@192.168.56.103's password:  
  
Number of key(s) added: 1  
  
Now try logging into the machine, with:  "ssh 'iquin@192.168.56.103'"  
and check to make sure that only the key(s) you wanted were added.  
  
iquin@Workstation:~$ ansible all -m apt -a update_cache=true --become --ask-beco  
me-pass  
BECOME password:  
192.168.56.103 | CHANGED => {  
    "ansible_facts": {  
        "discovered_interpreter_python": "/usr/bin/python3"  
    },  
    "cache_update_time": 1787126851,  
    "cache_updated": true,  
    "changed": true  
}
```

```
iquin@Workstation: ~
iquin@Workstation:~$ ansible all -m apt -a name=vim-nox --become --ask-become-pass
BECOME password:
192.168.56.103 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787126851,
  "cache_updated": false,
  "changed": true,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following package was automatically installed and is no longer required:\n libfwupd2\nUse 'sudo apt autoremove' to remove it.\nThe following additional packages will be installed:\n fonts-lato javascript-common libjs-jquery liblua5.1-0 libruby libruby3.2\n libsodium23 rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick\n ruby-xmlrpc ruby3.2 rubygems-integration vim-runtime\nSuggested packages:\n apache2 | lighttpd | httpd ri ruby-dev bundler cscope vim-doc\nThe following NEW packages will be installed:\n fonts-lato javascript-common libjs-jquery liblua5.1-0 libruby libruby3.2\n libsodium23 rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick\n ruby-xmlrpc ruby3.2 rubygems-integration vim-nox vim-runtime\n0 upgraded, 18 newly installed, 0 to remove and 1 not upgraded.\nNeed to get 18.6 MB of archives.\nAfter this operation
```

```
iquin@Workstation: ~
"Setting up ruby-sdbm:amd64 (1.0.0-5build4) ...",
"Setting up libruby3.2:amd64 (3.2.3-1ubuntu0.24.04.8) ...",
"Setting up ruby-rubygems (3.4.20-1) ...",
"Setting up vim-nox (2:9.1.0016-1ubuntu7.18) ...",
"update-alternatives: using /usr/bin/vim.nox to provide /usr/bin/ex (ex)
in auto mode",
"update-alternatives: using /usr/bin/vim.nox to provide /usr/bin/rview (r
view) in auto mode",
"update-alternatives: using /usr/bin/vim.nox to provide /usr/bin/rvim (r
vim) in auto mode",
"update-alternatives: using /usr/bin/vim.nox to provide /usr/bin/vi (vi)
in auto mode",
"update-alternatives: using /usr/bin/vim.nox to provide /usr/bin/view (v
view) in auto mode",
"update-alternatives: using /usr/bin/vim.nox to provide /usr/bin/vim (vi
vim) in auto mode",
"update-alternatives: using /usr/bin/vim.nox to provide /usr/bin/vimdiff
(vimdiff) in auto mode",
"Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...",
"Processing triggers for libc-bin (2.39-0ubuntu8.8) ...",
"Processing triggers for man-db (2.12.0-4build2) ..."

]
```

```
iquin@Workstation:~$ cd /var/log
iquin@Workstation:/var/log$ ls
alternatives.log      dmesg              kern.log
appport.log           dmesg.0            lastlog
apt                   dmesg.1.gz         openvpn
auth.log              dmesg.2.gz         private
boot.log              dmesg.3.gz         README
boot.log.1            dpkg.log            speech-dispatcher
bootstrap.log          faillog             sssd
btmptmp               fontconfig.log      syslog
cloud-init.log         gdm3                sysstat
cloud-init-output.log  gpu-manager.log     ubuntu-advantage-apt-hook.log
cups                   hp                  unattended-upgrades
cups-browsed           installer           vboxpostinstall.log
dist-upgrade           journal             wtmp
iquin@Workstation:/var/log$ cd apt
iquin@Workstation:/var/log/apt$ cat history.log
```

```
Start-Date: 2026-02-10 00:20:26
Commandline: apt-get -y --fix-policy install
Install: libgpg-error-l10n:amd64 (1.47-3build2, automatic), e2fsprogs-l10n:amd64
(1.47.0-2.4-exp1ubuntu4, automatic), libgpm2:amd64 (1.20.7-11, automatic), psmi
sc:amd64 (23.7-1build1, automatic), uuid-runtime:amd64 (2.39.3-9ubuntu6, automat
```

```

End Date: 2020-08-19 07:02:50
iquin@Workstation:/var/log/apt$ ansible all -m apt -a name=snapd --become --ask-
become-pass
BECOME password:
192.168.56.103 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787126851,
  "cache_updated": false,
  "changed": false
}
192.168.56.101 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787127129,
  "cache_updated": false,
  "changed": false
}
iquin@Workstation:/var/log/apt$

```

```

iquin@Workstation:/var/log/apt$ ansible all -m apt -a "name=snapd state=latest"
--become --ask-become-pass
BECOME password:
192.168.56.103 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787126851,
  "cache_updated": false,
  "changed": false
}
192.168.56.101 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787127129,
  "cache_updated": false,
  "changed": false
}
iquin@Workstation:/var/log/apt$

```

CREATING CPE232 & Playbook

```
iquin@Workstation:~/CPE232_Iquin$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
ok: [192.168.56.101]

TASK [install apache2 package] *****
ok: [192.168.56.101]
ok: [192.168.56.103]

PLAY RECAP *****
192.168.56.101      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.103      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

iquin@Workstation:~/CPE232_Iquin$
```

```
iquin@Workstation: ~/CPE232_Iquin
GNU nano 7.2 install_apache.yml
---
- hosts: all
  become: true
  tasks:
    - name: install apache2 package
      apt:
        name: apache2

[ Read 7 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
GNU nano 7.2                                install_apache.yml
--
- hosts: all
  become: true
  tasks:
    - name: install apache2 package
      apt:
        name: apache2xyz
```

```
iquin@Workstation: ~/CPE232_Iquin
iquin@Workstation:~/CPE232_Iquin$ nano install_apache.yml
iquin@Workstation:~/CPE232_Iquin$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.101]
ok: [192.168.56.103]

TASK [install apache2 package] *****
fatal: [192.168.56.103]: FAILED! => {"changed": false, "msg": "No package matching 'apache2xyz' is available"}
fatal: [192.168.56.101]: FAILED! => {"changed": false, "msg": "No package matching 'apache2xyz' is available"}

PLAY RECAP *****
192.168.56.101      : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.103      : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

iquin@Workstation:~/CPE232_Iquin$
```

```
iquin@Workstation:~/CPE232_Iquin$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.103]
ok: [192.168.56.101]

TASK [update repository index] *****
changed: [192.168.56.101]
changed: [192.168.56.103]

TASK [install apache2 package] *****
ok: [192.168.56.101]
ok: [192.168.56.103]

PLAY RECAP *****
192.168.56.101      : ok=3    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.103      : ok=3    changed=1    unreachable=0    failed=0    s
```

## PHP

```
iquin@Workstation: ~/CPE232_Iquin
iquin@Workstation:~/CPE232_Iquin$ nano install_apache.yml
iquin@Workstation:~/CPE232_Iquin$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.101]
ok: [192.168.56.103]

TASK [update repository index] *****
changed: [192.168.56.103]
changed: [192.168.56.101]

TASK [install apache2 package] *****
ok: [192.168.56.101]
ok: [192.168.56.103]

TASK [add PHP support for apache] *****
changed: [192.168.56.101]
changed: [192.168.56.103]

PLAY RECAP *****

TASK [install apache2 package] *****
ok: [192.168.56.101]
ok: [192.168.56.103]

TASK [add PHP support for apache] *****
changed: [192.168.56.101]
changed: [192.168.56.103]

PLAY RECAP *****
192.168.56.101      : ok=4    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.103      : ok=4    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

iquin@Workstation:~/CPE232_Iquin$
```

```
iquin@Workstation:~/CPE232_Iquin$ git add install_apache.yml
iquin@Workstation:~/CPE232_Iquin$ git commit -m "Activity 4: Task 2 - Ansible pl
aybook for apache2 and PHP"
[main 7d85990] Activity 4: Task 2 - Ansible playbook for apache2 and PHP
 1 file changed, 16 insertions(+)
 create mode 100644 install_apache.yml
iquin@Workstation:~/CPE232_Iquin$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 469 bytes | 469.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:MelquisedecIquin/CPE232_Iquin.git
 85a97d7..7d85990  main -> main
iquin@Workstation:~/CPE232_Iquin$
```


**CPE232\_Iquin**
Public
Pin
Watch 0

main
Go to file
+
<> Code


**MelquisedecIquin**
Activity 4: Task 2 - Ansible playbook for apache2 and ...
7d85990 · now

|                                                                                                        |                                           |                |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------|
|  README.md          | Initial commit                            | 26 minutes ago |
|  install_apache.yml | Activity 4: Task 2 - Ansible playbook ... | now            |

**README**


# CPE232\_Iquin

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of using a playbook?
  - A playbook lets you write down the exact state you want your servers to be in, then run it repeatedly and consistently across as many machines as you want. Instead of typing separate ad hoc commands every time, you write the steps once in a YAML file and reuse it anytime.
2. Summarize what we have done on this activity.
  - We started by running elevated ad hoc Ansible commands, testing that we need `--become` and `--ask-become-pass` to actually make changes on remote servers, since regular commands without privilege escalation failed. We used this to update the package cache, install VIM, and check/upgrade the snapd package.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

### Tools Used:

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

### Purpose of Use:

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

### Extent of Use:

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.